



Advanced Linear Algebra: Foundations to Frontiers

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1.4.2 Loss of digits of accuracy ¶

01.4.2 Digits of accuracy

fit width



Homework 1.4.2.1. Let $\alpha = -14.24123$ and $\hat{\alpha} = -14.24723$. Compute

- $|\alpha| =$
- $|\alpha - \hat{\alpha}| =$
- $\frac{|\alpha - \hat{\alpha}|}{|\alpha|} =$
- $\log_{10}\left(\frac{|\alpha - \hat{\alpha}|}{|\alpha|}\right) =$

▼ Solution

Let $\alpha = -14.24123$ and $\hat{\alpha} = -14.24723$. Compute

- $|\alpha| = 14.24123$
- $|\alpha - \hat{\alpha}| = 0.006$
- $\frac{|\alpha - \hat{\alpha}|}{|\alpha|} \approx 0.00042$
- $\log_{10}\left(\frac{|\alpha - \hat{\alpha}|}{|\alpha|}\right) \approx -3.4$

The point of this exercise is as follows:

- If you compare $\alpha = -14.24123$
 $\hat{\alpha} = -14.24723$ and you consider $\hat{\alpha}$ to be an approximation of α ,
then $\hat{\alpha}$ is accurate to four digits: -14.24 is accurate.
- Computing $\log_{10}\left(\frac{|\alpha - \hat{\alpha}|}{|\alpha|}\right)$ tells you approximately how many decimal
digits are accurate: 3.4 digits.

Be sure to read the solution to the last homework!